

Chapter 5

System and Memory

5.1 Overview

ESP32-C6 is an ultra-low power and highly-integrated system that integrates:

- a high-performance 32-bit RISC-V single-core processor (HP CPU), four-stage pipeline, clock frequency up to 160 MHz
- a low-power 32-bit RISC-V single-core processor (LP CPU), two-stage pipeline, clock frequency up to 20 MHz

All internal memory, external memory, and peripherals are located on the HP CPU and LP CPU buses.

5.2 Features

- **Address Space**
 - 832 KB of internal memory address space accessed from the instruction bus or data bus
 - 832 KB of peripheral address space
 - 16 MB of external memory virtual address space accessed from the instruction bus or the data bus
 - 512 KB of internal DMA address space
- **Internal Memory**
 - 320 KB internal ROM
 - 512 KB HP SRAM
 - 16 KB LP SRAM
- **External Memory**
 - Supports up to 16 MB external flash
- **Peripheral Space**
 - 51 modules/peripherals in total
- **GDMA**
 - 8 GDMA-supported modules/peripherals

Figure 5.2-1 illustrates the system structure and address mapping.

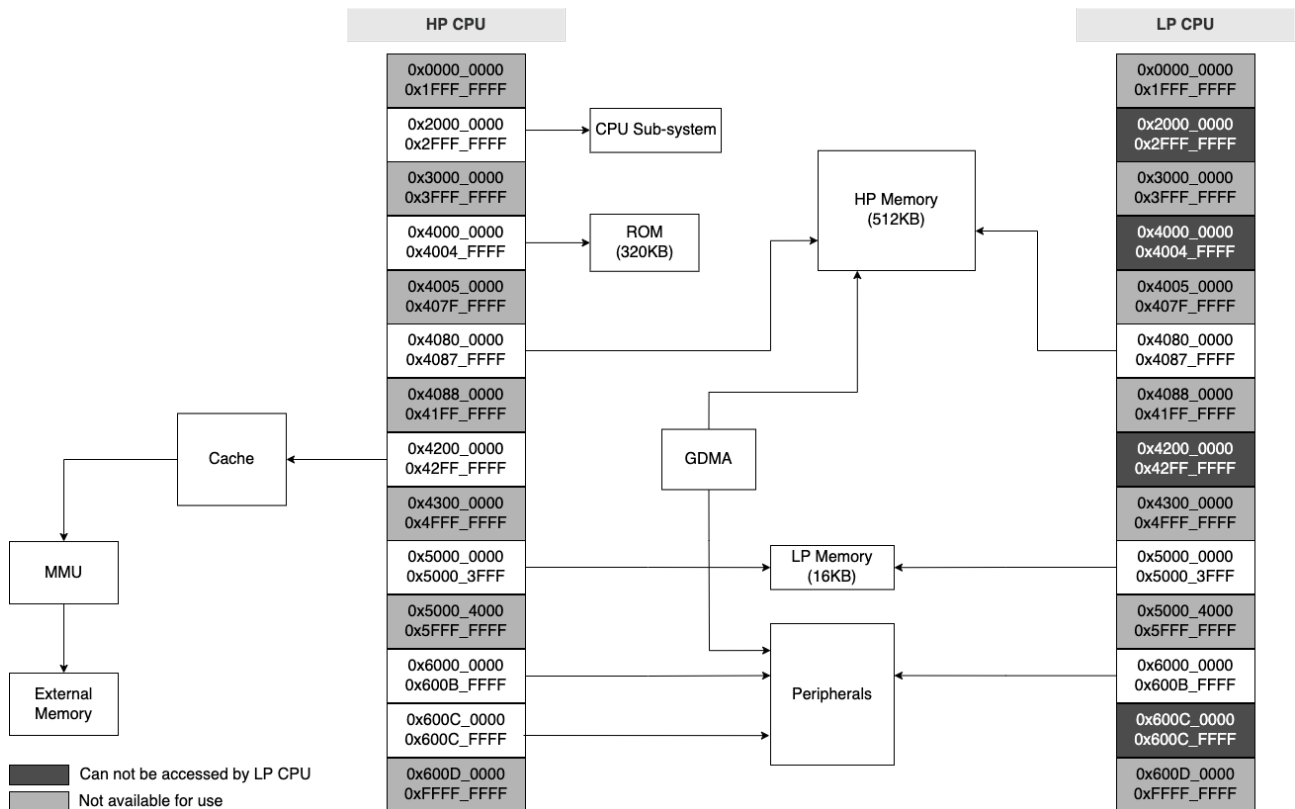


Figure 5.2-1. System Structure and Address Mapping

Note:

- The range of addresses available in the address space may be larger than the actual available memory of a particular type.
- For CPU Sub-system, please refer to Chapter 1 *High-Performance CPU*.

5.3 Functional Description

5.3.1 Address Mapping

All the non-reserved addresses are accessible by the instruction bus and the data bus, that is, the instruction bus and the data bus access the same address space.

Both data bus and instruction bus of the HP CPU and LP CPU are little-endian. The HP CPU and LP CPU can access data via the data bus using single-byte, double-byte, and 4-byte alignment.

The CPU can:

- directly access the internal memory via both data bus and instruction bus.
- (for HP CPU only) directly access the external memory which is mapped into the address space via cache.
- directly access modules/peripherals via data bus.

Table 5.3-1 lists the address ranges on the data bus and instruction bus and their corresponding target memories.

Table 5.3-1. Memory Address Mapping

Bus Type	Boundary Address		Size	Target
	Low Address	High Address		
	0x0000_0000	0x3FFF_FFFF		Reserved
Data/Instruction bus	0x4000_0000	0x4004_FFFF	320 KB	ROM [*]
	0x4005_0000	0x407F_FFFF		Reserved
Data/Instruction bus	0x4080_0000	0x4087_FFFF	512 KB	HP SRAM [*]
	0x4088_0000	0x41FF_FFFF		Reserved
Data/Instruction bus	0x4200_0000	0x42FF_FFFF	16 MB	External memory
	0x4300_0000	0x4FFF_FFFF		Reserved
Data/Instruction bus	0x5000_0000	0x5000_3FFF	16 KB	LP SRAM [*]
	0x5000_4000	0x5FFF_FFFF		Reserved
Data/Instruction bus	0x6000_0000	0x600C_FFFF	832 KB	Peripherals
	0x600D_0000	0xFFFF_FFFF		Reserved

^{*} All of the internal memories are managed by Permission Control module. An internal memory can only be accessed when it is allowed by Permission Control, then the internal memory can be available to the HP CPU and LP CPU. For more information about Permission Control, please refer to Chapter 16 [Permission Control \(PMS\)](#).

5.3.2 Internal Memory

ESP32-C6 consists of the following three types of internal memory:

- ROM (320 KB): The ROM is a read-only memory and can not be programmed. It contains the ROM code of some low-level system software and read-only data.
- HP SRAM (512 KB): The HP SRAM is a volatile memory that can be quickly accessed by the HP CPU or LP CPU (generally within a single HP CPU clock cycle for HP CPU).
- LP SRAM (16 KB): LP SRAM is also a volatile memory, however, in Deep-sleep mode, data stored in the LP SRAM will not be lost. The LP SRAM can be accessed by the HP CPU or LP CPU and is usually used to store program instructions and data that need to be kept in sleep mode.

1. ROM

This 320 KB ROM is a read-only memory, addressed by the HP CPU through the instruction bus or through the data bus via 0x4000_0000 ~ 0x4004_FFFF, as shown in Table 5.3-1.

2. HP SRAM

This 512 KB HP SRAM is a read-and-write memory, accessed by the HP CPU or LP CPU through the instruction bus or through the data bus as shown in Table 5.3-1.

3. LP SRAM

This 16 KB LP SRAM is a read-and-write memory, accessed by the HP CPU or LP CPU through the instruction bus or through the data bus via their shared address 0x5000_0000 ~ 0x5000_3FFF as shown in Table 5.3-1.

LP SRAM can be accessed by the following modes:

- high-speed mode, i.e., the LP SRAM is accessed in HP CPU clock frequency. In this case:
 - HP CPU can access the LP SRAM without any latency.
 - But the latency of LP CPU accessing LP SRAM ranges from a few dozen to dozens of LP CPU cycles.
- low-speed mode, i.e., the LP SRAM is accessed in LP CPU clock frequency. In this case:
 - LP CPU can access the LP SRAM with zero cycle latency.
 - But the latency of HP CPU accessing LP SRAM ranges from a few dozen to dozens of HP CPU cycles.

You can switch the modes based on your application scenarios.

- If the LP CPU is not working, you can switch to high-speed mode to improve the access speed of the HP CPU.
- If the LP CPU is executing code in the LP SRAM, you can switch to the low-speed mode.

When the HP CPU is in sleep mode, you must switch to the low-speed mode.

Detailed configuration is as follows:

- Configure [LP_AON_FAST_MEM_MUX_SEL](#) to select the mode needed:
 - 1: high-speed mode
 - 0: low-speed mode
- Set [LP_AON_FAST_MEM_MUX_SEL_UPDATE](#) to start mode switch.
- Read [LP_AON_FAST_MEM_MUX_SEL_STATUS](#) to check if mode switch is done:
 - 0: mode is switched
 - 1: mode is not switched

5.3.3 External Memory

ESP32-C6 supports SPI, Dual SPI, Quad SPI, and QPI interfaces that allow connection to external flash.

ESP32-C6 also supports hardware manual encryption and automatic decryption based on XTS-AES algorithm to protect users' programs and data in the external flash.

5.3.3.1 External Memory Address Mapping

The HP CPU accesses the external memory via the cache. According to information inside the MMU (Memory Management Unit), the cache maps the HP CPU's address (0x4200_0000 ~ 0x42FF_FFFF) into a physical address of the external memory. Due to this address mapping, ESP32-C6 can address up to 16 MB external flash. Note that the instruction bus shares the same address space (16 MB) with the data bus to access the external memory.

5.3.3.2 Cache

As shown in Figure 5.3-1, ESP32-C6 has a read-only uniform cache which is four-way set-associative. Its size is 32 KB and its block size is 32 bytes. The cache is accessible by the instruction bus and the data bus at the same time, but can only respond to one of them at a time. When a cache miss occurs, the cache controller will initiate a request to the external memory.

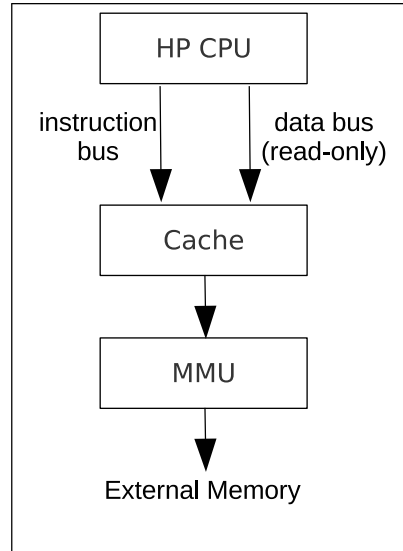


Figure 5.3-1. Cache Structure

5.3.3.3 Cache Operations

ESP32-C6 cache supports the following operations:

1. **Invalidate:** This operation is used to remove valid data in the cache. Once this operation is done, the deleted data is stored only in the external memory. If the HP CPU wants to access the data again, it needs to access the external memory. There are two types of invalidate operation: Invalidate-All and Manual-Invalidate. Manual-Invalidate is performed only on data in the specified area in the cache, while Invalidate-All is performed on all data in the cache.
2. **Preload:** This operation is to load instructions and data into the cache in advance. The minimum unit of preload-operation is one block. There are two types of preload-operation: manual preload (Manual-Preload) and automatic preload (Auto-Preload). Manual-Preload means that the hardware prefetches a piece of continuous data according to the virtual address specified by the software. Auto-Preload means the hardware prefetches a piece of continuous data according to the current address where the cache hits or misses (depending on configuration).
3. **Lock/Unlock:** The lock operation is used to prevent the data in the cache from being easily replaced. There are two types of lock: prelock and manual lock. When prelock is enabled, the cache locks the data in the specified area when filling the missing data to cache memory, while the data outside the specified area will not be locked. When manual lock is enabled, the cache checks the data that is already in the cache memory and locks the data only if it falls in the specified area, and leaves the data outside the specified area unlocked. When there are missing data, the cache will replace the data in the unlocked way first, so the data in the locked way is always stored in the cache and will not be replaced. But when all ways within the cache are locked, the cache will replace data, as if it was not locked. Unlocking is the reverse of locking, except that it only can be done manually.

Please note that Manual-Invalidate operation only works on the unlocked data. If you expect to perform such operation on the locked data, please unlock them first.

5.3.4 GDMA Address Space

The General Direct Memory Access (GDMA) peripheral consisting of three TX channels and three RX channels provides Direct Memory Access (DMA) service, including:

- data transfers between different locations of internal memory
- data transfers between modules/peripherals and internal memory

GDMA uses the same addresses as the data bus to access HP SRAM, i.e., GDMA uses address range 0x4080_0000 ~ 0x4087_FFFF to access HP SRAM.

Eight modules/peripherals in ESP32-C6 work together with GDMA. As shown in Figure 5.3-2, eight vertical lines correspond to these eight modules/peripherals with GDMA function. The horizontal line represents a certain channel of GDMA (can be any channel), and the intersection of the vertical line and the horizontal line indicates that a module/peripheral has the ability to access the corresponding channel of GDMA. If there are multiple intersections on the same line, it means that these peripherals/modules can not enable the GDMA function at the same time.

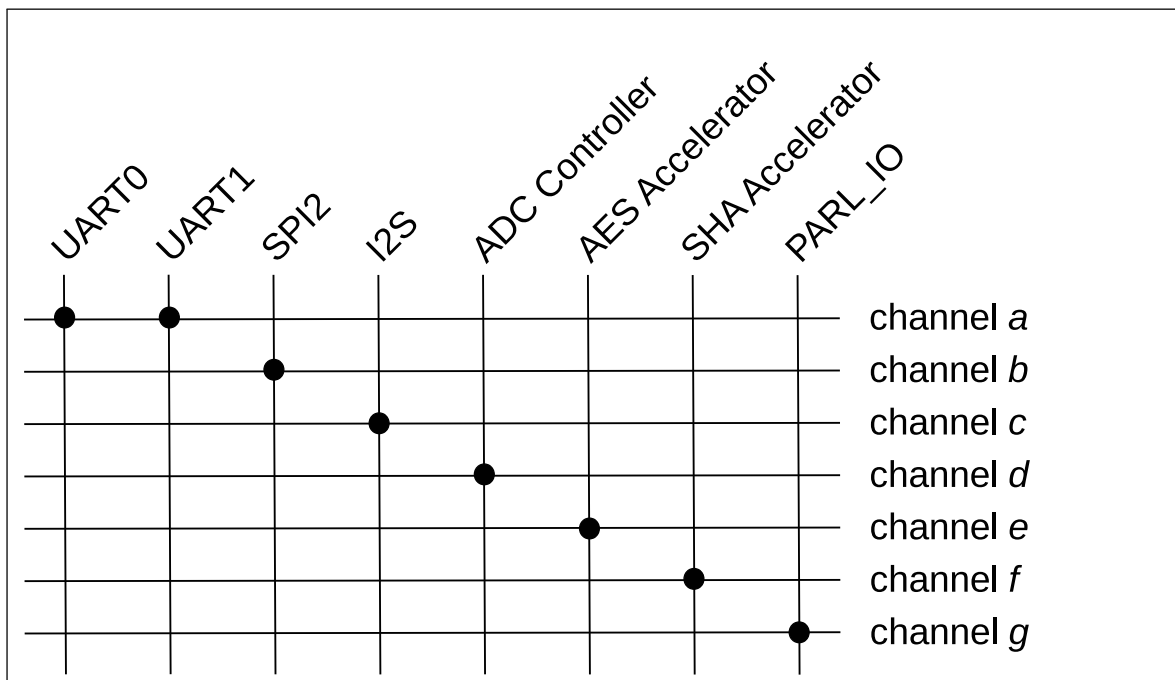


Figure 5.3-2. Modules/peripherals that can work with GDMA

These modules/peripherals can access any memory available to GDMA. For more information, please refer to Chapter 4 [GDMA Controller \(GDMA\)](#).

Note:

When accessing a memory via GDMA, a corresponding access permission is needed, otherwise this access may fail. For more information about permission control, please refer to Chapter 16 [Permission Control \(PMS\)](#).

5.3.5 Modules/Peripherals Address Mapping

Table 5.3-2 lists all the modules/peripherals and their respective address ranges. Note that the address space of specific modules/peripherals is defined by “Boundary Address” (including both Low Address and High Address).

Table 5.3-2. Module/Peripheral Address Mapping

Target	Boundary Address		Size (KB)
	Low Address	High Address	
UART Controller 0 (UART0)	0x6000_0000	0x6000_0FFF	4
UART Controller 1 (UART1)	0x6000_1000	0x6000_1FFF	4
External Memory Encryption and Decryption (XTS_AES)	0x6000_2000	0x6000_2FFF	4
Reserved	0x6000_3000	0x6000_3FFF	
I2C Controller (I2C)	0x6000_4000	0x6000_4FFF	4
UHCI Controller (UHCI)	0x6000_5000	0x6000_5FFF	4
Remote Control Peripheral (RMT)	0x6000_6000	0x6000_6FFF	4
LED PWM Controller (LEDC)	0x6000_7000	0x6000_7FFF	4
Timer Group 0 (TIMG0)	0x6000_8000	0x6000_8FFF	4
Timer Group 1 (TIMG1)	0x6000_9000	0x6000_9FFF	4
System Timer (SYSTIMER)	0x6000_A000	0x6000_AFFF	4
Two-wire Automotive Interface 0 (TWAIO)	0x6000_B000	0x6000_BFFF	4
I2S Controller (I2S)	0x6000_C000	0x6000_CFFF	4
Two-wire Automotive Interface 1 (TWA1)	0x6000_D000	0x6000_DFFF	4
Successive Approximation ADC (SAR ADC)	0x6000_E000	0x6000_EFFF	4
USB Serial/JTAG Controller	0x6000_F000	0x6000_FFFF	4
Interrupt Matrix (INTMTX)	0x6001_0000	0x6001_0FFF	4
Reserved	0x6001_1000	0x6001_1FFF	
Pulse Count Controller (PCNT)	0x6001_2000	0x6001_2FFF	4
Event Task Matrix (SOC_ETM)	0x6001_3000	0x6001_3FFF	4
Motor Control PWM (MCPWM)	0x6001_4000	0x6001_4FFF	4
Parallel IO Controller (PARL_IO)	0x6001_5000	0x6001_5FFF	4
SDIO HINF*	0x6001_6000	0x6001_6FFF	4
SDIO SLC*	0x6001_7000	0x6001_7FFF	4
SDIO SLCHOST*	0x6001_8000	0x6001_8FFF	4
Reserved	0x6001_9000	0x6007_FFFF	
GDMA Controller (GDMA)	0x6008_0000	0x6008_0FFF	4
General Purpose SPI2 (GP-SPI2)	0x6008_1000	0x6008_1FFF	4
Reserved	0x6008_2000	0x6008_7FFF	
AES Accelerator (AES)	0x6008_8000	0x6008_8FFF	4
SHA Accelerator (SHA)	0x6008_9000	0x6008_9FFF	4
RSA Accelerator (RSA)	0x6008_A000	0x6008_AFFF	4
ECC Accelerator (ECC)	0x6008_B000	0x6008_BFFF	4
RSA Digital Signature Peripheral (RSA_DS)	0x6008_C000	0x6008_CFFF	4

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Target	Boundary Address		Size (KB)
	Low Address	High Address	
HMAC Accelerator (HMAC)	0x6008_D000	0x6008_DFFF	4
Reserved	0x6008_E000	0x6008_FFFF	
IO MUX	0x6009_0000	0x6009_0FFF	4
GPIO Matrix	0x6009_1000	0x6009_1FFF	4
Memory Access Monitor (MEM_MONITOR)*	0x6009_2000	0x6009_2FFF	4
Reserved	0x6009_4000	0x6009_4FFF	
HP System Register (HP_SYSREG)	0x6009_5000	0x6009_5FFF	4
Power/Clock/Reset (PCR) Register	0x6009_6000	0x6009_6FFF	4
Reserved	0x6009_7000	0x6009_7FFF	
Trusted Execution Environment (TEE) Register*	0x6009_8000	0x6009_8FFF	4
Access Permission Management Controller (HP_APM)*	0x6009_9000	0x6009_9FFF	4
Reserved	0x6009_A000	0x600A_FFFF	
Power Management Unit (PMU)	0x600B_0000	0x600B_03FF	1
Low-power Clock/Reset Register (LP_CLKRST)	0x600B_0400	0x600B_07FF	1
eFuse Controller (EFUSE)	0x600B_0800	0x600B_0BFF	1
RTC Timer (RTC_TIMER)	0x600B_0C00	0x600B_0FFF	1
Low-power Always-on Register (LP_AON)	0x600B_1000	0x600B_13FF	1
Low-power UART (LP_UART)	0x600B_1400	0x600B_17FF	1
Low-power I2C (LP_I2C)	0x600B_1800	0x600B_1BFF	1
RTC Watch Dog Timer (RTC_WDT)	0x600B_1C00	0x600B_1FFF	1
Low-power IO MUX (LP IO MUX)	0x600B_2000	0x600B_23FF	1
I2C Analog Master (I2C_ANA_MST)	0x600B_2400	0x600B_27FF	1
Low-power Peripheral (LPPERI)	0x600B_2800	0x600B_2BFF	1
Low-power Analog Peripheral (LP_ANA_PERI)	0x600B_2C00	0x600B_2FFF	1
Reserved	0x600B_3000	0x600B_33FF	
Low-power Trusted Execution Environment (LP_TEE)*	0x600B_3400	0x600B_37FF	1
Low-power Access Permission Management (LP_APM)*	0x600B_3800	0x600B_3BFF	1
Reserved	0x600B_3C00	0x600B_FFFF	
RISC-V Trace Encoder (TRACE)	0x600C_0000	0x600C_0FFF	4
Reserved	0x600C_1000	0x600C_1FFF	
DEBUG ASSIST (ASSIST_DEBUG)*	0x600C_2000	0x600C_2FFF	4
Reserved	0x600C_3000	0x600C_4FFF	
Interrupt Priority Register (INTPRI)	0x600C_5000	0x600C_5FFF	4
Reserved	0x600C_6000	0x600C_FFFF	

* The address space of this module/peripheral is not continuous.

Note:

As shown in the figure [5.2-1](#)

- HP CPU can access all peripherals listed in the table [5.3-2](#).
- LP CPU can access all peripherals listed in the table [5.3-2](#) except [RISC-V Trace Encoder \(TRACE\)](#), [DEBUG ASSIST \(ASSIST_DEBUG\)](#) and [Interrupt Priority Register \(INTPRI\)](#).